

1. Importamos librerias

```
!pip install ydata-profiling
import numpy as np
import pandas as pd
#from pandas.tools.plotting import scatter_matrix
pd.set_option('display.width', 500)
pd.set_option('display.max_columns', 100)
pd.set_option('display.notebook_repr_html', True)
import seaborn as sns
sns.set(style="whitegrid")
import warnings
warnings.filterwarnings('ignore')
import string
import math
import sys
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import OneHotEncoder
import sklearn
from IPython.core.interactiveshell import InteractiveShell
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from ydata_profiling import ProfileReport

InteractiveShell.ast_node_interactivity = "all"
```

[Mostrar el resultado oculto](#)

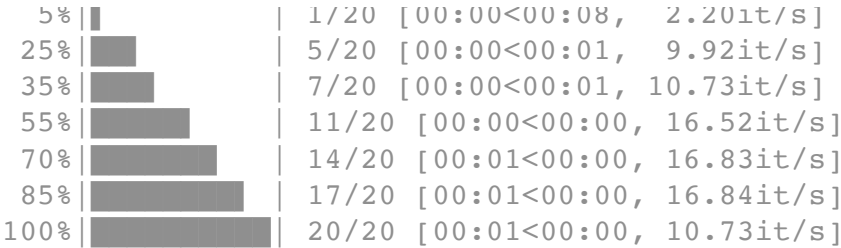
```
df = pd.read_csv("student_salary.csv")

profile = ProfileReport(df, title="Reporte de análisis exploratorio",
profile.to_notebook_iframe())
```

Summarize dataset: 100%

79/79 [00:27<00:00, 2.97it/s, Completed]

0% | | 0/20 [00:00<?, ?it/s]



Generate report structure: 100%

1/1 [00:09<00:00, 9.92s/it]

Render HTML: 100%

1/1 [00:00<00:00, 1.70it/s]

Reporte de análisis exploratorio

Memory size

439.6 KiB

More details

company_type

Categorical

High correlation

Missing

Distinct	4	MNC	3179
Distinct (%)	0.1%	Startup	2230
Missing	1298	Mid-size	2158
Missing (%)	14.4%	Top Tech	135

Memory size

482.1 KiB

More details

job_role

Categorical

High correlation

Missing

Distinct

4

Web De...

1939

Distinct (%)

0.1%

Data Sci...

1938

```

# Leer archivo
df = pd.read_csv("student_salary.csv")

print(df.columns)

# Eliminar filas con valores vacios
df = df.dropna()

# Eliminar columna que no aporta información relevante
df = df.drop("student_id", axis=1)

# Separar variables independientes y variable objetivo
X = df.drop("salary_lpa", axis=1)
y = df["salary_lpa"]

# Separar en entrenamiento y prueba
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
)

# Columnas categóricas
categorical_features = ["branch", "company_type", "job_role"]

# Rellenar valores nulos en columnas categóricas
for feature in categorical_features:
    X_train[feature] = X_train[feature].fillna("Unknown")
    X_test[feature] = X_test[feature].fillna("Unknown")

# Aplicar One Hot Encoding a train y test
X_train = pd.get_dummies(X_train, columns=categorical_features, dtype=int)
X_test = pd.get_dummies(X_test, columns=categorical_features, dtype=int)

# Alinear columnas para que train y test tengan las mismas columnas
X_train, X_test = X_train.align(X_test, join="left", axis=1, fill_value=0)

```

```
# Escalamiento
sc = StandardScaler()

X_train_scaled = sc.fit_transform(X_train)
X_test_scaled = sc.transform(X_test)

print(X_train)
print(X_test)
```

```
Index(['student_id', 'cgpa', 'branch', 'college_tier', 'python_skill',
      'dsa_skill', 'ml_skill', 'web_dev_skill'],
      dtype='object')
cgpa college_tier python_skill dsa_skill ml_skill web_dev_skill
8201 8.52 2 1 0 1
7977 8.32 1 1 0 0
5589 8.41 1 1 1 0
5675 6.86 2 1 1 0
5503 7.95 3 0 0 1
...
6940 7.12 2 1 1 0
214 6.74 2 1 1 1
4702 8.88 3 1 1 0
7295 9.54 2 1 1 1
5339 5.30 1 1 1 0
```

```
[6161 rows x 29 columns]
cgpa college_tier python_skill dsa_skill ml_skill web_dev_skill
5639 8.07 3 1 1 1
1883 6.34 1 1 0 1
1438 9.44 3 0 1 1
6043 7.15 3 1 1 1
7601 9.54 1 1 0 0
...
4961 5.54 1 0 0 0
3684 5.29 3 1 1 0
5271 8.04 3 0 1 0
5828 9.16 1 1 0 0
3459 7.58 1 0 0 0
```

```
[1541 rows x 29 columns]
```

